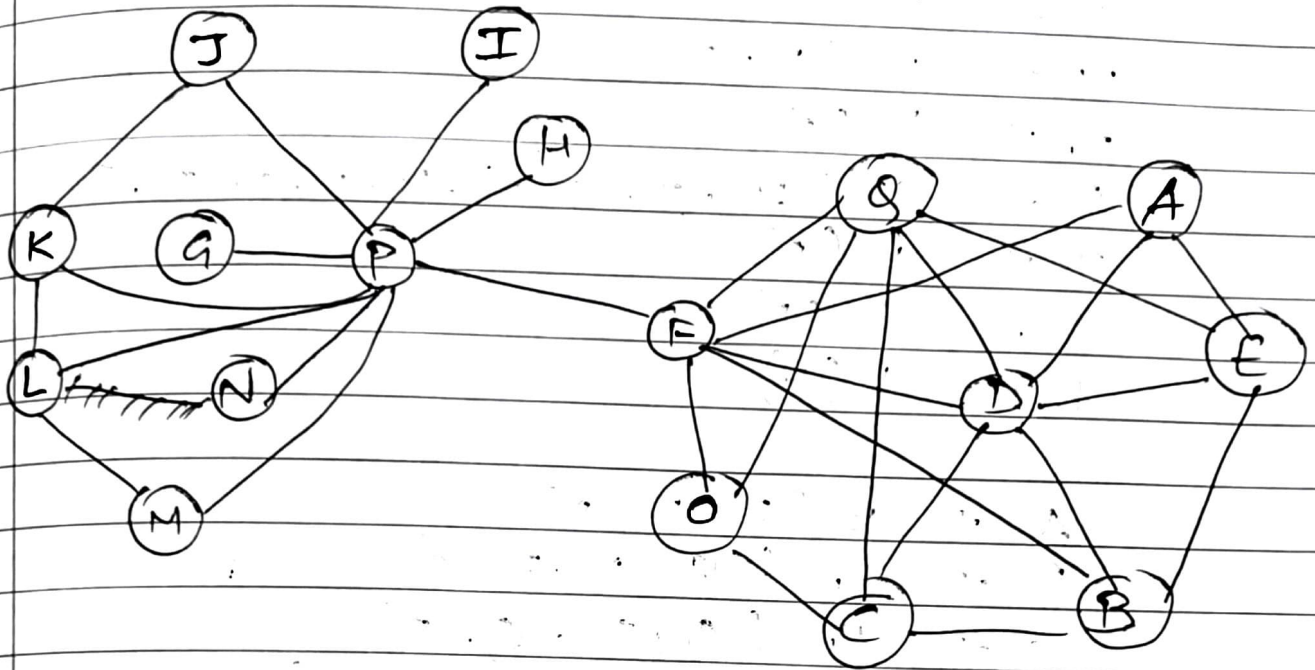


2. Social Network Structure, Measures & Visualization

Q.1 How degree distribution is plotted for the graph? Show degree distribution of the following graph.

⇒



i)

- Degree is used to describe individual nodes.
- To get an idea of the degree for all the nodes in the network, we can build the degree distribution.
- This shows how many nodes have each possible degree.
- It involves counting degree of each node and then the next step is to count how many nodes have each degree.
- There is a formula to calculate density:

$$\frac{\text{number of edges}}{\text{number of possible edges}}$$

2) Calculate Degree of each node:

J $\rightarrow$ 2 (P, K)

K $\rightarrow$ 3 (J, L, P)

L $\rightarrow$ 3 (K, M, P)

M $\rightarrow$ 2 (L, P)

N $\rightarrow$ 1 (P)

P $\rightarrow$ 9 (H, I, J, G, K, L, N, M, E)

I $\rightarrow$ 1 (P)

G $\rightarrow$ 1 (P)

H $\rightarrow$ 1 (P)

A $\rightarrow$ 3 (F, E, D)

B $\rightarrow$ 4 (E, D, F, C)

C $\rightarrow$ 4 (O, D, Q, B)

D $\rightarrow$ 5 (Q, A, E, B, C)

E $\rightarrow$ 4 (A, D, B, Q)

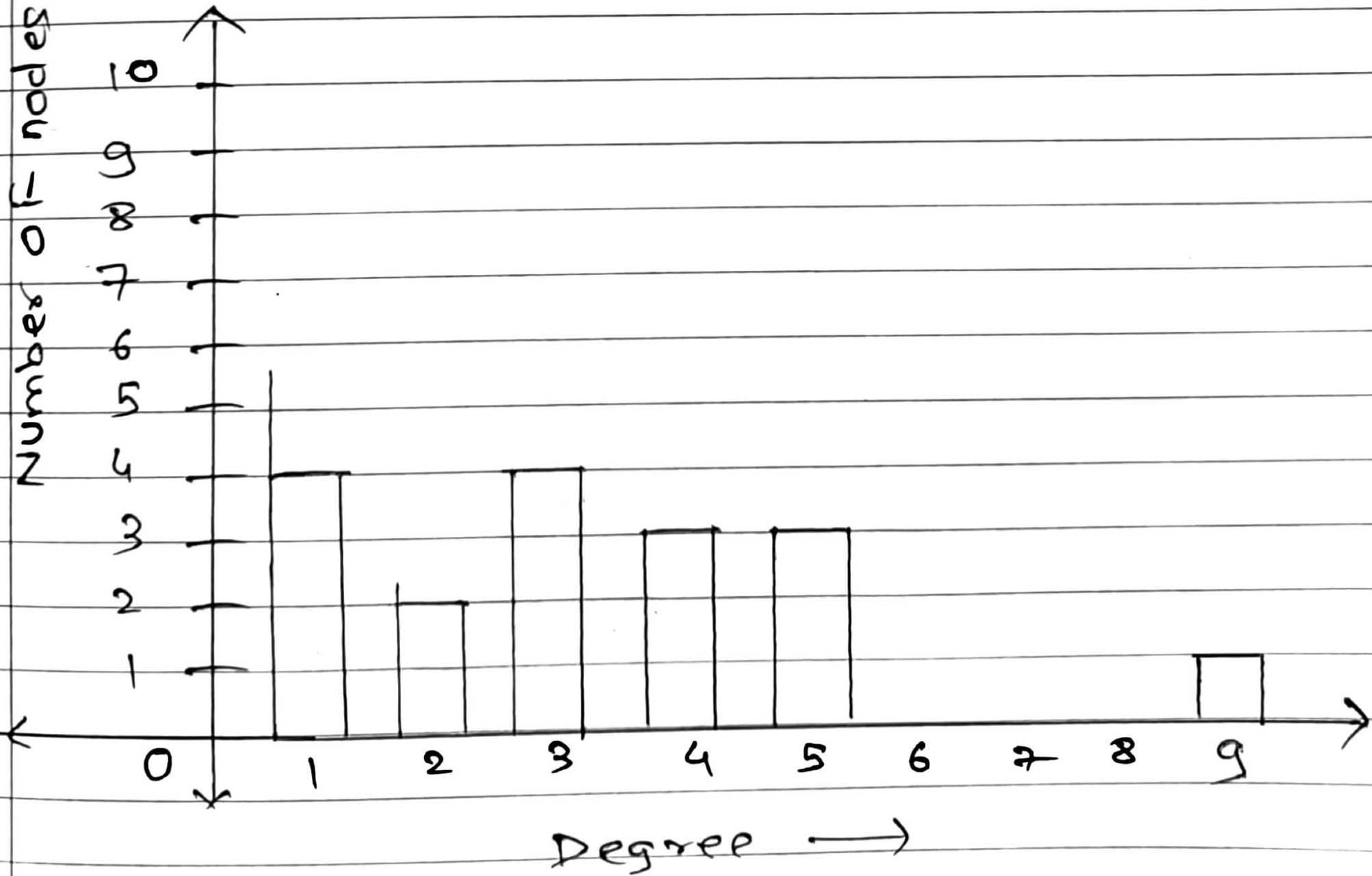
F $\rightarrow$ 5 (A, B, Q, O, P)

O $\rightarrow$ 3 (F, Q, C)

Q $\rightarrow$ 5 (F, E, O, C, D)

3) Count how many nodes have each degree:

Degree	Count
1	4 (N, I, G, H)
2	2 (J, M)
3	4 (K, L, A, O)
4	3 (B, C, E)
5	3 (D, F, Q)
9	1 (P)



Q.2 Centralization in social media analytics with example.

⇒

- Centralization is an important way to understand the role of a node in the network and to compare nodes.
- Centralization uses distribution of a centrality measure to understand the network as a whole.
- Since there are different centrality measures there are different centralization measures of a graph.
- Centralization is computed by looking at the sum of differences in centrality between the most central node and every other node in the network, and dividing this by the maximum possible differences in centrality that could exist in the graph.
- Let $C(n)$ be the centrality of node n , using whatever centrality measures we choose.
- Say n is the most central node. We want to find the difference in centrality between n and every other node in the network, and add those up.
- If there are N nodes in the network the formula for this is:

$$\sum_{i=1}^N C(n^*) - C(n_i)$$

- Then, we want to divide this by the sum of the maximum possible differences between n and every other node.

- However, this maximum possible centrality will change depending on which centrality measures we are using.
- Denote this using the formula will max in front.

$$\max \sum_{i=1}^N C(n^*) \cdot C(n_i)$$

- Now, we compute centralization.
- It is equal to the sum of differences divided by maximum possible sum of differences:

$$\sum_{i=1}^N C(n^*) \cdot C(n_i)$$

$$\max \sum_{i=1}^N C(n^*) \cdot C(n_i)$$

- Types of Centrality :-

- 1) Degree centrality
- 2) Betweenness centrality
- 3) Closeness centrality
- 4) Eigenvector centrality

1) Degree centrality

⇒ It counts the number of direct connections a node has in the network.

- In-degree: No. of incoming links.
- Out-degree: No. of outgoing links.
- Formula:

For a node v :

$$C_D(v) = \deg(v)$$

- For e.g.,

A Facebook user with many friends.

2) Betweenness centrality

⇒ Betweenness centrality measures the no. of times a node lies on the shortest path between other nodes.

- In directed networks, betweenness can have a different meaning.
- A user with high betweenness may be followed by many others who don't follow the same people as the user.
- Betweenness centrality can in fact be used in optimizing City Transport planning.
- The vertex betweenness centrality $BC(v)$ of a vertex v is defined as follows:-

$$BC(v) = \sum_{u, w \in V} \left(\frac{\sigma_{uw}(v)}{\sigma_{uw}} \right)$$

σ_{uw} = total no. of shortest path b/w u and w

$\sigma_{uw}(v)$ = total no. of shortest path b/w u and w through v

3) Closeness centrality

⇒ Closeness centrality indicates how close a node is to all other nodes in the network.

- It is calculated as the average of the shortest path length from the node to every other node in the network.
- Formula:

$$C_c(v) = \frac{1}{\sum_t d(v,t)}$$

4) Eigenvector centrality

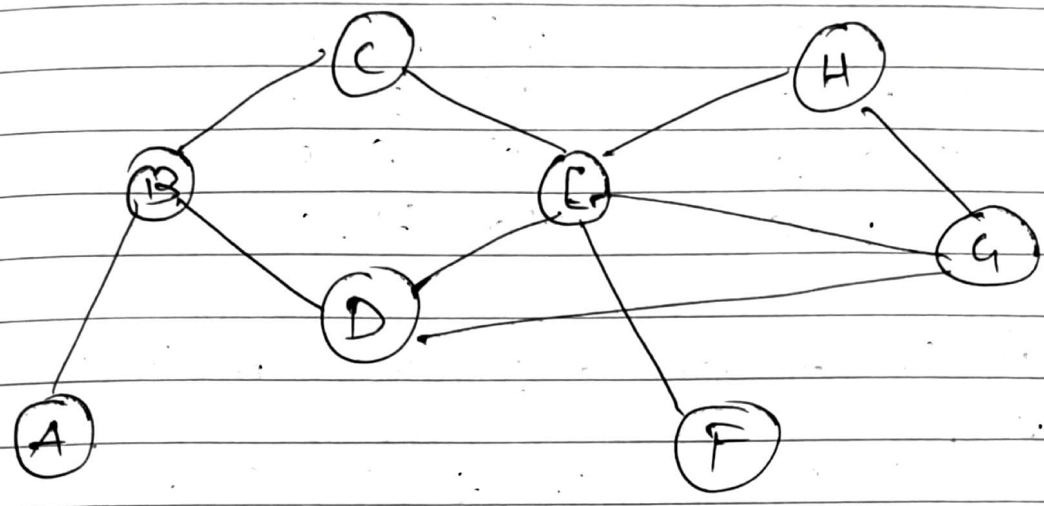
⇒ Eigenvector centrality measures a node's importance while giving consideration to the importance of its neighbors.

- It is determined by performing a matrix calculation to determine what is called the principal eigenvector using the adjacency matrix.
- Eigenvector centrality is at the core of Google's PageRank algorithm, which ~~uses~~ they use to rank web pages.
- Formula:

It solves the equation -

$$Ax = \lambda x$$

Q.3 Answer the following questions about a graph:



⇒ i) How many nodes are in the network?
 ⇒ There are 8 nodes in the graph,
 i.e. A, B, C, D, E, F, G, H

ii) How many edges are in the network?
 ⇒ There are 10 edges in the graph,
 i.e.

- | | |
|-------|-------|
| A - B | E - H |
| B - C | E - G |
| B - D | E - F |
| C - E | H - G |
| D - E | |
| D - G | |

iii) Is this graph directed or undirected?
 ⇒ since the all connections are mutual and there's no direction arrow!
 Undirected.

iv) Create an adjacency list for this graph
 ⇒

- | | |
|---------|----------|
| 1) A, B | 6) D, G |
| 2) B, D | 7) E, H |
| 3) B, C | 8) E, G |
| 4) C, E | 9) E, F |
| 5) D, E | 10) H, G |

v) Create an adjacency matrix

⇒

	A	B	C	D	E	F	G	H
A	0	1	0	0	0	0	0	0
B	1	0	1	1	0	0	0	0
C	0	1	0	0	1	0	0	0
D	0	1	0	0	1	0	1	0
E	0	0	1	1	0	1	1	1
F	0	0	0	0	1	0	0	0
G	0	0	0	1	1	0	0	1
H	0	0	0	0	1	0	1	0

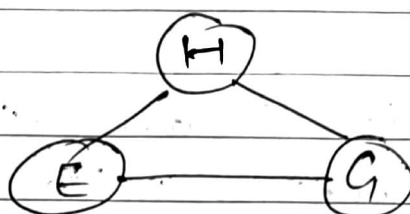
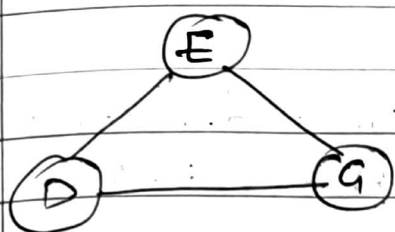
vi) What is length of shortest path from node A to node F?

⇒ There are two paths of same length 4.

- a) A - B - D - E - F
- b) A - B - C - E - F

vii) What is the largest clique in the network? How many cliques of that size are there?

⇒ A clique is a fully-connected subgraph.
- Only 2 cliques of size 3 exists.



viii) How many connected components are there in this graph?

⇒ The graph is fully connected.

∴ No. of connected components = 1

ix) Estimate the density of the graph!

⇒

$$\therefore \text{Density} = \frac{2 \times \text{no. of edges}}{n(n-1)}$$

$$= \frac{2 \times 10}{8 \times 7}$$

$$= 0.3571$$

x) Are there any hubs in the network?

IF so which nodes and why it is a hub?

⇒ Hubs are the most connected nodes in the network.

- Yes, there are hub in the network.

- Node E is a hub in the network.
- Node E is most connected with 5 edges to it, which is highest compared to other nodes in the network.

Q.4 Explain Degree distribution, density of connections, tie strengths, Hub.

⇒

1) Degree Distribution

⇒ The degree of a node is the number of edges connected to it.

- Degree distribution shows how degree are distributed across the network.
- It helps identify the structure of the network, whether it's centralized or decentralized.

2) Density of connections

⇒ The ratio of the actual number of connections to the total possible connections between all the nodes.

- Formula:

$$\text{Density} = \frac{2 \times \text{no. of edges}}{n(n-1)}$$

- Measures how tightly-knit the network is.

3) Tie Strength

⇒ Represents the intensity or closeness of the relationship between two nodes.

- Types :-

a) Strong ties : Close friend, frequent interactions.

b) Weak ties : Acquaintances, occasional interactions.

- Weak ties often bridge different communities and spread information widely.

4) Hub

⇒ A hub is a highly connected node in a network with a high degree.

- Plays a key role in information dissemination.

- Influences network flow and viral marketing.

Q.5 Social Media Network Analysis.

⇒

- Networks are the building blocks of social media and can provide valuable business insights.
- Social media network analysis therefore deals with the structure, analysis and understanding of social media networks.

Common Social Media Network Types:

1) Friendship networks

⇒ Friendship network let people maintain social ties and share content with people they closely associate with, such as family and friends.

2) Follower networks

⇒ Followers are the people who follow you on social media platform.

3) Fan Network

⇒ A fan network is formed by social media fans or supporters of someone or something, such as product, service, person, etc.

4) Group network

⇒ Group networks are formed by people who share common interests and agendas.

5) Professional Networks

⇒ LinkedIn is a good example of professional networks where people manage their professional identity by creating a profile that lists their achievements, education, work history, etc.

• Types of Networks

1) Based on Existence

⇒ There are two types -

a) Implicit Networks: Implicit networks do not exist by default and need to be intentionally constructed with the help of dedicated tools and techniques.

For e.g., Keyword co-occurrence network.

b) Explicit Networks: Explicit network exists by default; in other words, they are explicitly designed for social media users to be part of.

E.g., Facebook friendship network.

2) Based on Directions

⇒ There are two types -

a) Directed Networks: A network with directed links among nodes is called a directed network.

For e.g., Twitter follow-follow network

b) Undirected Network: In this, the links among the nodes do not have any direction. For e.g., A Facebook friendship network

3) Based on Mode

⇒ There are two types:-

- a) One-Mode : A one-mode network is formed among a single set of nodes of the same nature.
- b) Two-Mode : A bimodal network is a network with two sets of nodes of different classes.

4) Based on Weights

⇒ There are two types -

- a) Weighted : In this, the links among nodes bear certain weights to indicate the strength of association among the nodes.
- b) Unweighted : In this, links among nodes does not bear weights.

• Tools

- 1) NodexL : NodexL is a free social network analysis and visualization tool.
- 2) UCINET : UCINET is a social network analysis software application for windows operating system.
- 3) Pajek : Pajek is a software application for analyzing and visualizing large networks.

Q.6 Describe density, bridge and hub of a social network with example.

⇒

1) Density

⇒ Density describes how interconnected the nodes in the network are.

- It is calculated as the ratio of actual connections (edges) to all possible connections among the nodes.

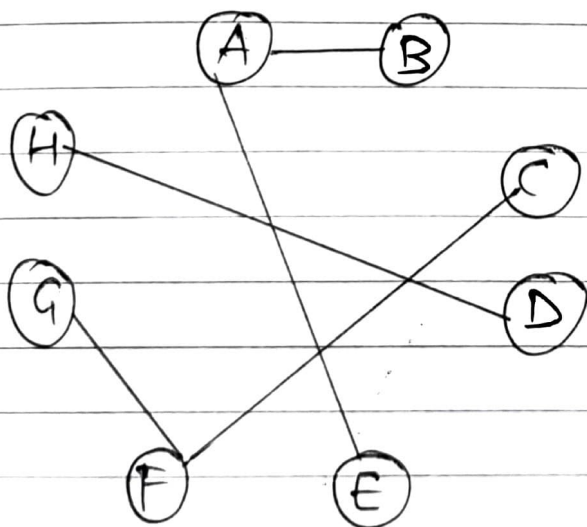
- Formula:

$$\text{Density} = \frac{\text{No. of edges}}{\text{No. of possible edges}}$$

- For an undirected graph/network of n nodes:

$$\text{Possible connections} = \frac{n(n-1)}{2}$$

- For e.g.,



$$\begin{aligned} \text{Density} &= \frac{5}{\left(\frac{8 \times 7}{2}\right)} \\ &= \frac{10}{56} \\ &= 0.179 \end{aligned}$$

2) Bridge

⇒ A bridge is a connection or node that connects two otherwise disconnected parts of a network.

- Removing it would increase the number of disconnected component.
- Bridge help in Information diffusion across communities.
- They are critical points for maintaining network connectivity.

3) Hub

⇒ A hub is a node with a high degree. It plays a central role in spreading information or influencing others in the network.

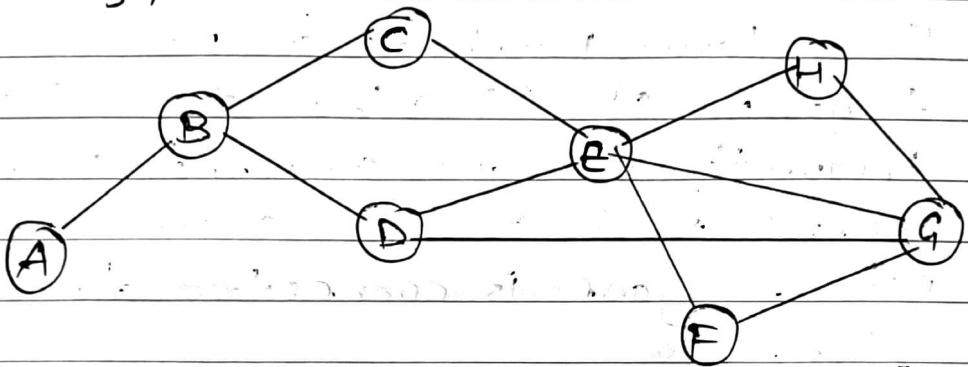
- Plays a key role in information dissemination.
- Influences network Flow and viral marketing.

Q.7 Define clique and cluster with example.

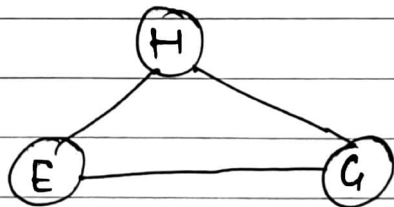
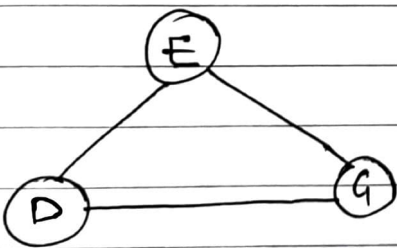
⇒

1) Clique :

- A clique is a subset of nodes in a network where every node is directly connected to every other node in the subset.
- It ~~completes~~ forms a complete subgraph.
- For e.g.,



- Cliques in above graph :-



- Often represents strongly connected and closely collaborating groups.

2) Cluster

⇒ A cluster is a group of nodes in which the nodes are more densely connected internally than they are externally with the rest of the network.

- Used in community detection to analyze group behavior, interests, etc.
- Can contain multiple cliques.
- For e.g.;

In a large Facebook group -

A group of 15 college classmates who frequently tag and comment on each other's posts forms a cluster, even if not everyone is connected to everyone else.

Q.8 List and explain different network layouts.
⇒

- Network Layouts

- 1) Graph layout
- 2) Random layout
- 3) circular layout
- 4) Grid layout
- 5) Force-directed layout

1) Graph layout

⇒ The graph layouts dictate the placements of nodes and edges.

- Researchers have presented some general guidelines that make network visualizations easier to work with:

- i) Every node is visible
- ii) For every node you can count its degree.
- iii) For every link you can follow it from sources to destination.
- iv) Clusters and outliers are identified.

- They start with the nodes randomly placed and iteratively moves them around into better positions.

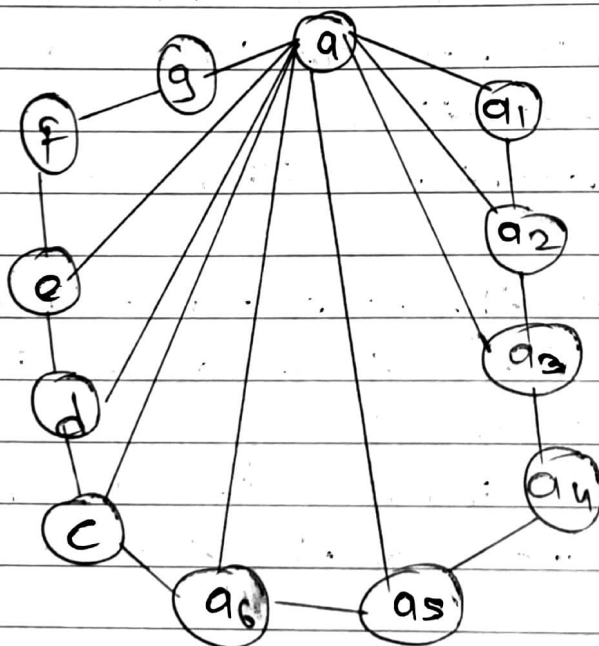
- As a result, running the algorithm multiple times will produce graphs that look different.

2) Random layout

- ⇒ Often, when loading data into a visualization tool, the nodes are placed randomly.
- This is called a random layout, and it often does not provide much insights into the structure of the network.
 - We may be able to tell that node a has a high degree in this network, but the clusters and other patterns are not at all clear from the random layout.

3) Circular layout

- ⇒ Circular layouts place all the nodes in a circle and then add edges between them.
- Some circular layouts place nodes closer to one another when they are more closely connected.



- A circular layouts places nodes in structured positions and then adds edges between connected pairs.
- Another way to do this is to place nodes in a grid.

4) Grid layout

- ⇒ Nodes are placed in a grid-like structures, similar to a matrix.
- Useful for comparing nodes attributes, works well with bipartite graphs.
- Visualizing connections between two distinct nodes sets.

5) Force-directed layout

- ⇒ Most graphs are not laid out randomly or in one of these formats with a predetermined structure.
- Instead, the layout is dynamic and determined by the connections between the nodes.
- Those nodes that are more closely connected are laid out close to one another, and those that are distant are shown further apart.
- Nodes and edges are treated as a physical system, and a simulation of that system is applied to determine a final layout.

Q.9 What are different scale issues that occur in network visualization?

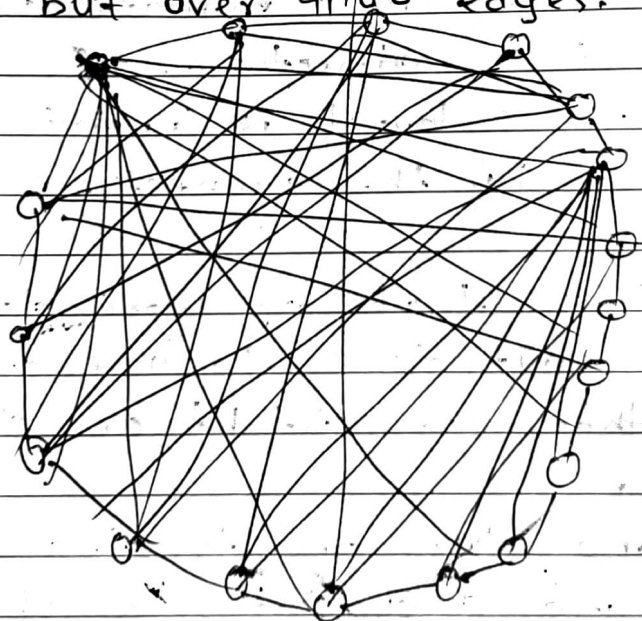
⇒

- Visualization is very useful for analysing networks of this size or smaller.
- When networks become much larger, the quality of the visualization diminishes.
- Nodes represent hosts and edges represent connections between them.

1) Density

⇒ Density can also be a problem for visualization, even if the number of nodes is small.

- The Figure shows a network of members of the U.S. Senate. There are only 100 nodes but over 4,000 edges.



- The edges indicate that the senators have voted the same way in at least one bill.

2) Filtering for visual patterns

⇒ It is often difficult to see any patterns in very dense networks.

- One way to compensate for this is to filter the networks when possible.
- For example if we take ~~from the same~~ network

3) Visual Scalability

⇒ Visualization becomes less effective on small screens or low-resolution displays when networks are large.

- Implement responsive design, collapsible views, and multi-level exploration.

4) Loss of Interpretability

⇒ As the network grows, patterns, relationships and key insights becomes difficult to extract visually.

- Use summary metrics and community detection to guide visualization.

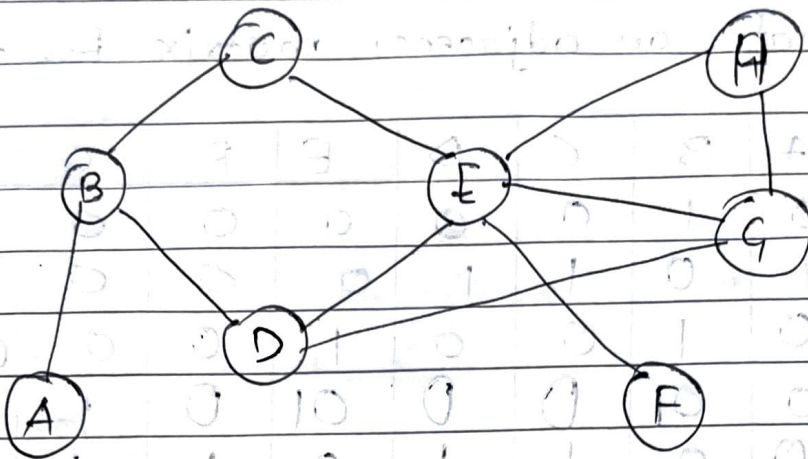
5) Layout degradation

⇒ Layout algorithms struggle to maintain clarity as the network size increases.

- Switch to layout algorithms optimized for large-scale graphs.

6) Dynamic scale problems

- ⇒ In temporal networks, scale issues arise when multiple time slices need to be visualized or animated.
- Use timeline-based or interactive filtering approaches.



a) How many nodes are in the network?
 $\Rightarrow$ There are 8 nodes: $\{A, B, C, D, E, F, G, H\}$

b) How many edges are in the network?
 $\Rightarrow$ 10

c) Is this graph directed or undirected?
 $\Rightarrow$ This graph is undirected because edges do not have arrows indicating direction.

d) Create an adjacency list for this graph.

$\Rightarrow$

A $\rightarrow$ B

B $\rightarrow$ A, C, D

C $\rightarrow$ B, E

D $\rightarrow$ B, E, G

E $\rightarrow$ C, G, H, D, F

F $\rightarrow$ E

G $\rightarrow$ E, H, D

H $\rightarrow$ E, G

1) A, B

2) B, D

3) B, C

4) C, E

5) D, E

6) D, G

7) E, H

8) E, G

9) E, F

10) H, G

e) Create an adjacency matrix for this graph

	A	B	C	D	E	F	G	H
A	0	1	0	0	0	0	0	0
B	1	0	1	1	0	0	0	0
C	0	1	0	0	1	0	0	0
D	0	0	0	0	0	0	1	0
E	0	0	1	1	0	1	1	1
F	0	0	0	0	1	0	0	0
G	0	0	0	1	1	0	0	1
H	0	0	0	0	1	0	1	0

f) What is the length of shortest path from node A to F

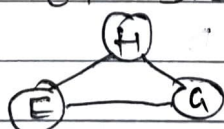
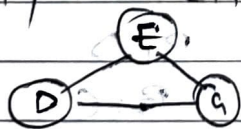
⇒

1) Path: A → B → D → E → F 2) A → B → C → E → F

Length: 4

g) What is the largest clique in this network?
How many cliques of that size are there?

⇒ The largest clique is {D, E, G}, {H, E, G}
Only 2 cliques of size 3 exists.

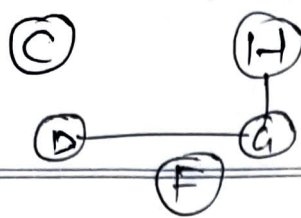


h) How many connected components are there in this network?

⇒ The graph is fully connected.

No. of connected components = 1
= {A, B, C, D, E, F, G, H}

fig: 1.5 egocentric network of node E with E excluded



Singletons = C, H

Page No.:

Date:

i) Draw the 1.5 ego network for node E
How many singletons are in the ego network?

⇒ If we want to see only E's neighbours and their conditions, it is called a 1.5 degree egocentric network;

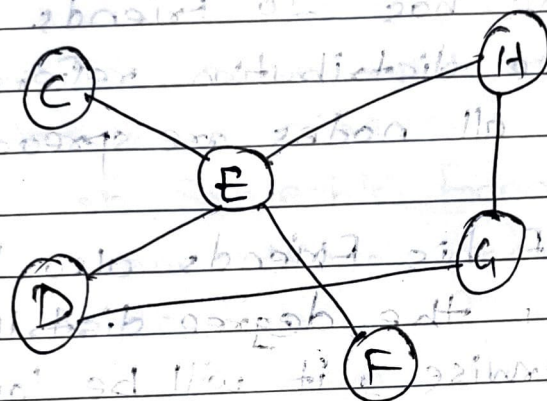


Fig: 1.5 degree egocentric network of E

j) Are there any hubs in network? If so, which nodes and why it is hub?

⇒ Hubs are most connected nodes in the network.

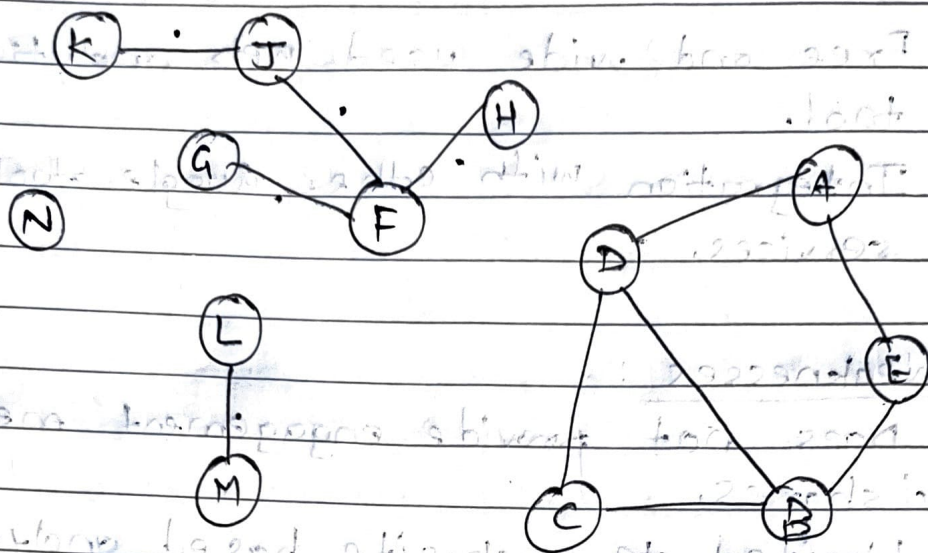
- Yes, there are hub in the network;

- Node E is a hub in the network;

- Node E is most connected with 5 edges to it, which is highest compare to other nodes in the network.

Consider the graph

(I)



a) How many singletons are there in the network? List them

⇒ A singleton is a node that is not connected to any other node.

- From the graph, the singleton nodes are:

i) I

ii) N

- Thus, total no. of singletons = 2

b) What is the largest connected component?

⇒ A connected component is a subgraph in which every node is connected to at least one other node.

- From the graph, the two major connected components are:

Component 1 : {A, B, C, D, E}

Component 2 : {F, G, H, J, K}

c) Are there any bridges in the network?

IF- so, where are they?

⇒ A bridge is an edge that, if removed would disconnect a part of the graph.

- Bridges in the graph :-

F-H

F-G

F-B

B-K

L-M

d) Create an adjacency list for the network.

⇒

1) A, D

2) A, E

3) B, E

4) B, D

5) B, C

6) D, C

7) L, M

8) K, J

9) J, F

10) F, G

11) F, H

12)

e) Create an adjacency matrix :-

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
A	0	0	0	1	1	0	0	0	0	0	0	0	0	0
B	0	0	1	1	1	0	0	0	0	0	0	0	0	0
C	0	1	0	1	0	0	0	0	0	0	0	0	0	0
D	1	1	1	0	0	0	0	0	0	0	0	0	0	0
E	1	1	0	0	0	0	0	0	0	0	0	0	0	0
F	0	0	0	0	0	0	1	1	0	1	0	0	0	0
G	0	0	0	0	0	1	0	0	0	0	0	0	0	0
H	0	0	0	0	0	1	0	0	0	0	0	0	0	0
I	0	0	0	0	0	0	0	0	0	0	0	0	0	0
J	0	0	0	0	0	0	0	0	0	1	0	0	0	0
K	0	0	0	0	0	0	0	0	0	1	0	0	0	0
L	0	0	0	0	0	0	0	0	0	0	0	0	1	0
M	0	0	0	0	0	0	0	0	0	0	0	1	0	0
N	0	0	0	0	0	0	0	0	0	0	0	0	0	0

1. 1 1 1 1 1 1 1 1 1 1 1 1 1 1

2. 1 1 1 1 1 1 1 1 1 1 1 1 1 1

3. 1 1 1 1 1 1 1 1 1 1 1 1 1 1

4. 1 1 1 1 1 1 1 1 1 1 1 1 1 1

5. 1 1 1 1 1 1 1 1 1 1 1 1 1 1

6. 1 1 1 1 1 1 1 1 1 1 1 1 1 1

7. 1 1 1 1 1 1 1 1 1 1 1 1 1 1

8. 1 1 1 1 1 1 1 1 1 1 1 1 1 1

9. 1 1 1 1 1 1 1 1 1 1 1 1 1 1

10. 1 1 1 1 1 1 1 1 1 1 1 1 1 1